

Descriptive statistics/plots continued
and data transformations

Overview

Review

- Plots and statistics of categorical data
- Measures of central tendency

Continuation of statistics and plots of quantitative data

- Measures of spread
- Two quantitative variables

Start on data transformations using dplyr

Reminder: Homework 2

It is due on Gradescope by 11pm on Monday July 13th

- Question 4 involves reading a short article and commented on it, so you can get started on this right away

How is the homework going so far?



Review: Categorical data

Categorical variables take on one of a fixed number of possible values

For categorical variables we usually want to view:

- **Frequency table:** How many items are each category or
- **Relative frequency table:** The proportion (or percentage) of items in each category

Vector of drinking behavior

```
> drinking_vec <- profiles$drinks
```

Frequency and relative frequency tables

```
> drinks_table <- table(drinking_vec)
```

```
> prop.table(drinks_table)
```

	age	body_type	diet	drinks	drugs	education
1	22	a little extra	strictly anything	socially	never	working on college/university
2	35	average	mostly other	often	sometimes	working on space camp
3	38	thin	anything	socially	NA	graduated from masters program
4	23	thin	vegetarian	socially	NA	working on college/university
5	29	athletic	NA	socially	never	graduated from college/university
6	29	average	mostly anything	socially	NA	graduated from college/university

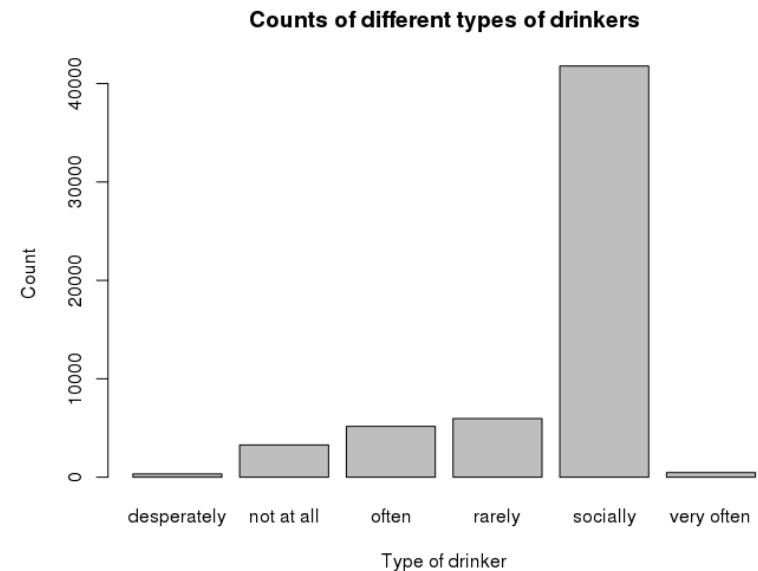
Review: Visualizing categorical data

We can plot the number of items in each category using a bar plot

```
barplot(drinks_table,  
        ylab = "Count",  
        xlab = "Type of drinker")
```

We can also use the `pie()` function to create pie charts

```
pie(drinks_table)
```

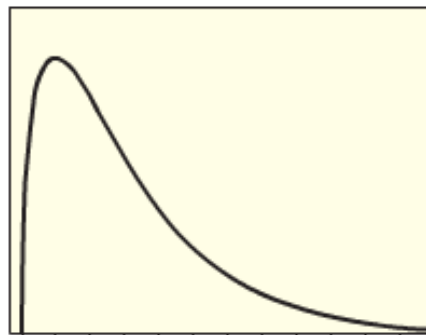


Review Visualizing quantitative data

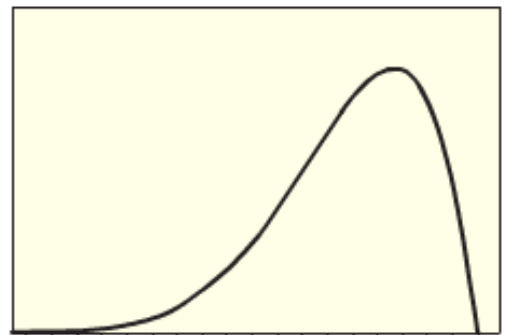
We can visualize quantitative data using histograms

```
hist(profiles$height, breaks = 50)
```

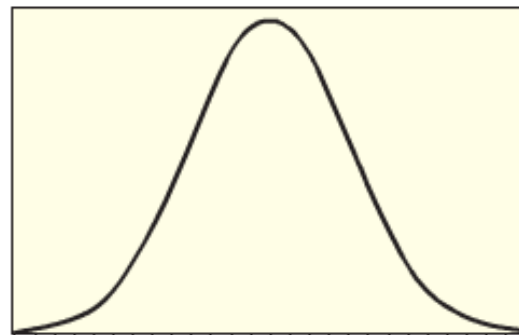
Common shapes of histograms are:



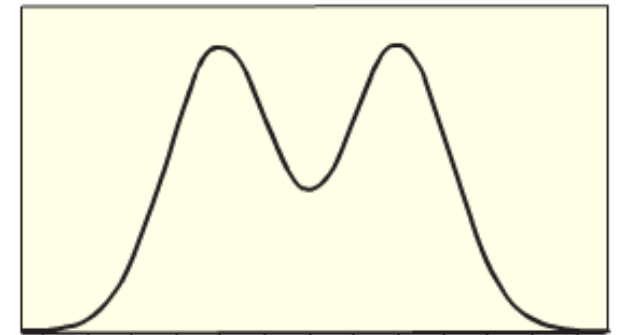
(a) Skewed to the right



(b) Skewed to the left



(c) Symmetric and bell-shaped



(d) Symmetric but not bell-shaped

Review Measures of central tendency

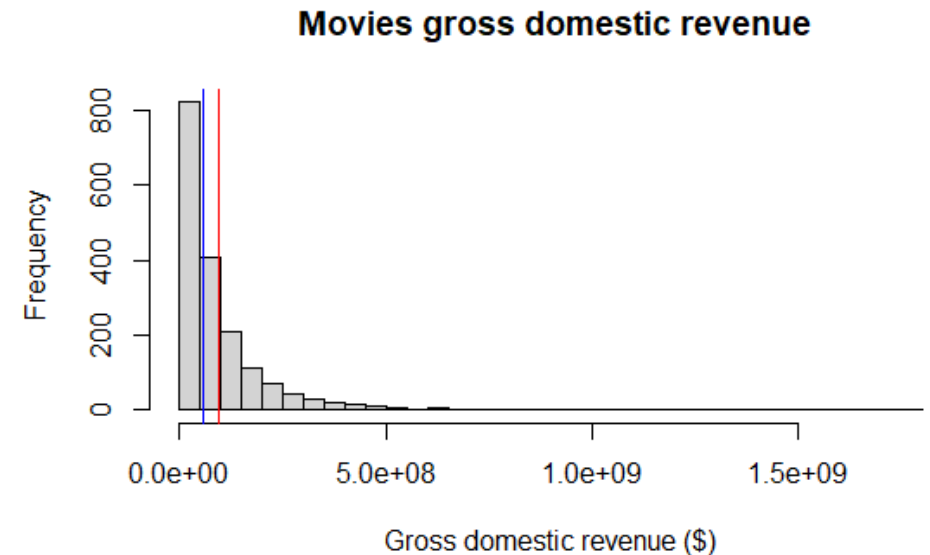
One common measure of central tendency is the **mean**

$$\text{mean}(x, \text{na.rm} = \text{TRUE}) \quad \frac{1}{n} \sum_{i=1}^n x_i$$

The **median** is the value such that half of the data is less than the median and half are greater than the median

$$\text{median}(v, \text{na.rm} = \text{TRUE})$$

The median is resistant to extreme values while the mean is not



Example:

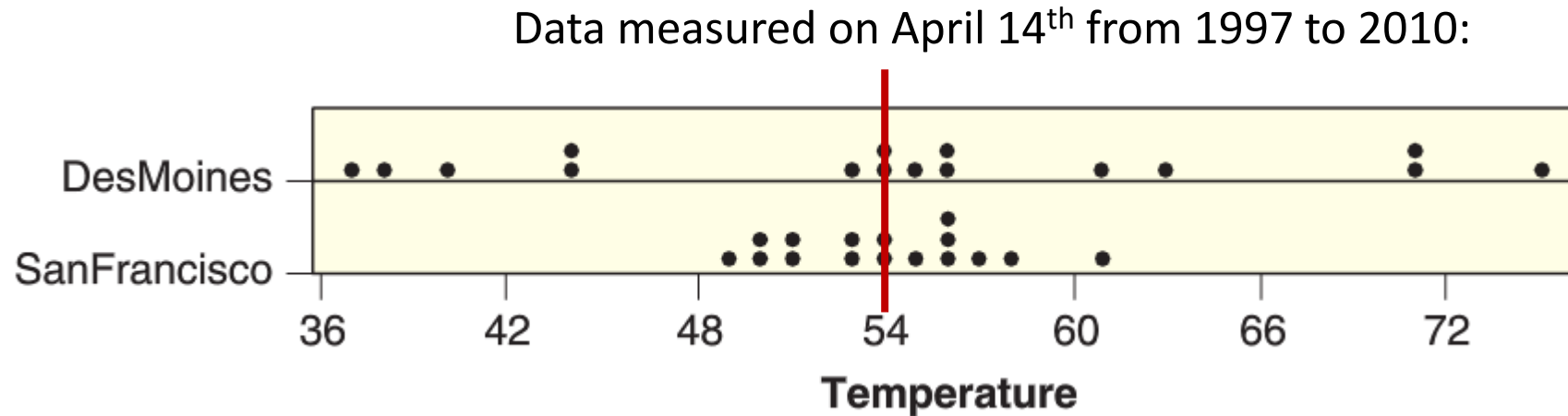
Mean US salary = \$72,641

Median US salary = \$51,939

Measures of spread

Measure of spread 1: standard deviation

The **standard deviation** is a statistic that quantifies how far the data is spread



Mean temperature (°F):	Des Moines = 54.49	San Francisco = 54.01
Standard deviation (°F):	Des Moines = 11.73	San Francisco = 3.38

Example: computing the standard deviation

Suppose we had a sample with $n = 4$ points:

$$x_1 = 8, \quad x_2 = 2, \quad x_3 = 6, \quad x_4 = 4,$$

We can compute the mean using the formula:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{4} \cdot (x_1 + x_2 + x_3 + x_4) = \frac{1}{4} \cdot (8 + 2 + 6 + 4) = 5$$

The standard deviation can be computed using the formula:

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

(remember order of operations!)

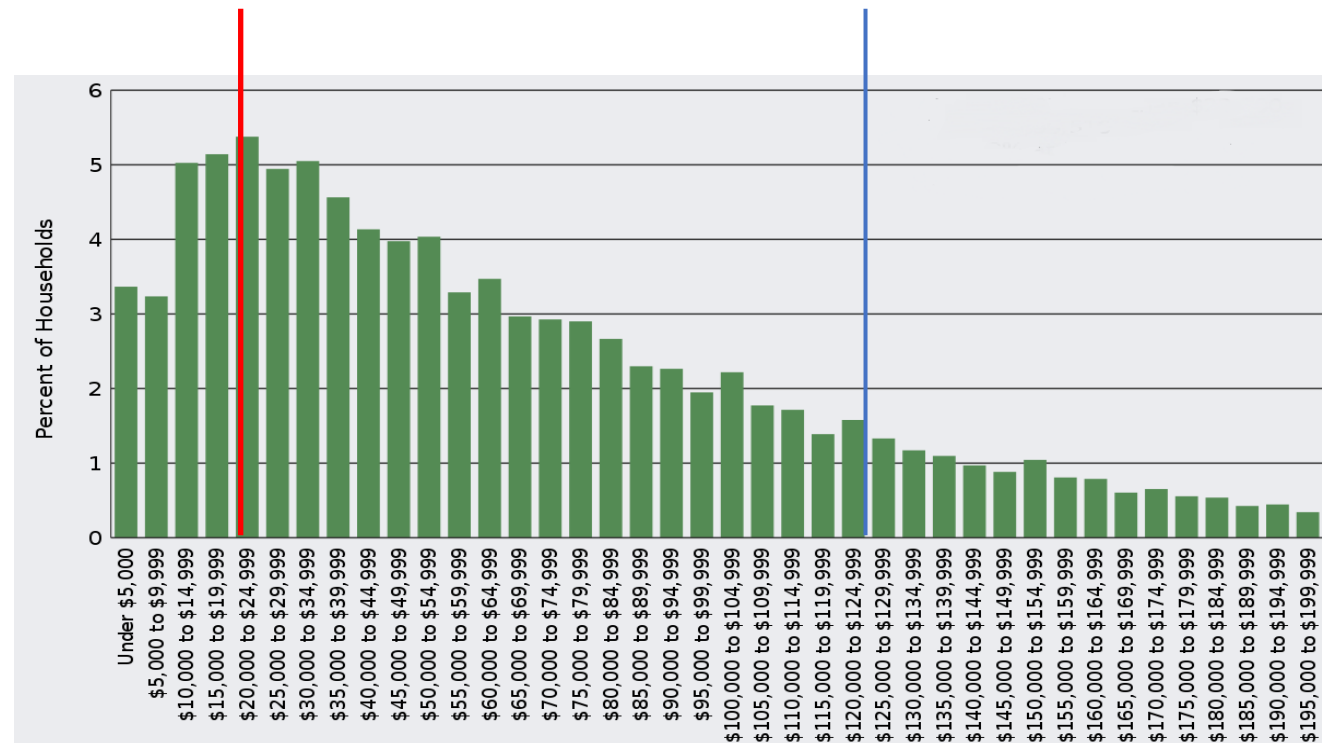
Percentiles

The **Pth percentile** is the value of a quantitative variable which is greater than P percent of the data

For the US income distribution what are the 20th and 80th percentiles?

20th percentile = \$21,430

80th percentile = \$112,254



R: `quantile(v, .8)`

Five Number Summary

A **five-number summary** is a set of five descriptive statistics that provides a concise overview of a dataset's distribution

Five Number Summary = (minimum, Q_1 , median, Q_3 , maximum)

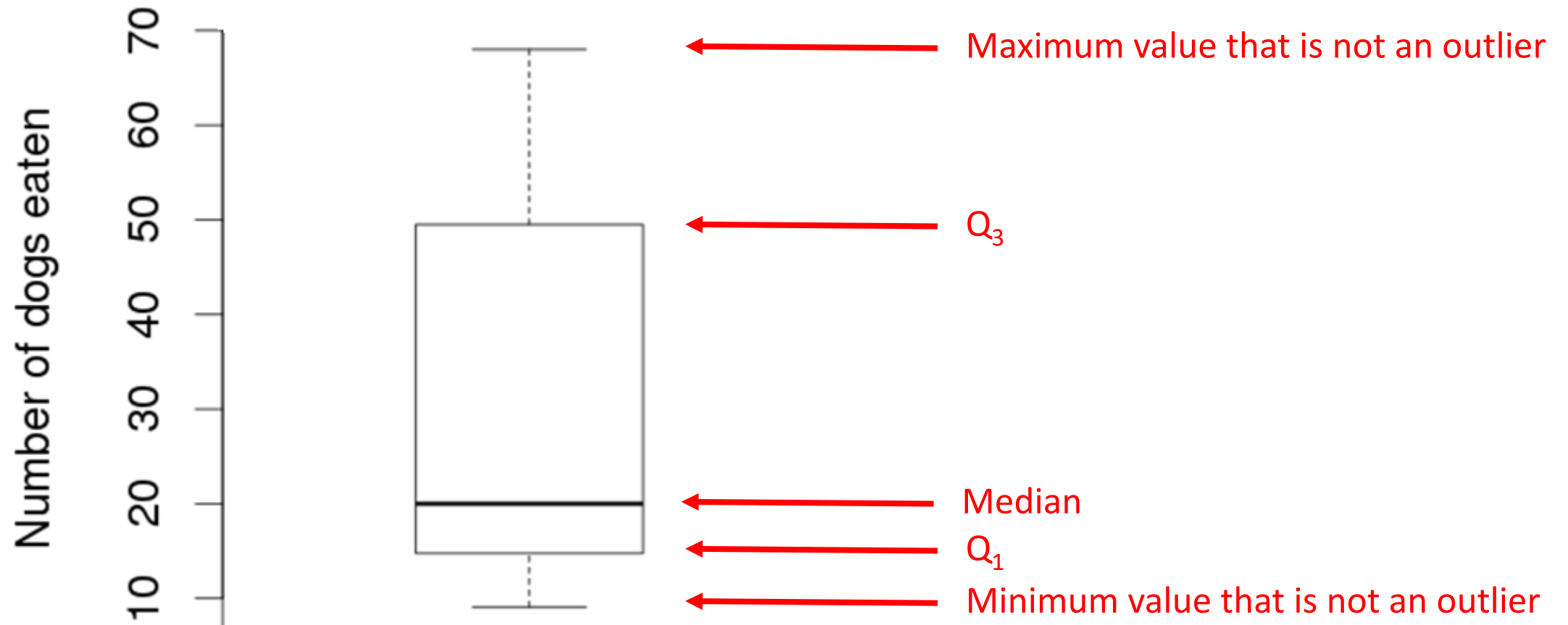
Q_1 = 25th percentile (also called 1st quartile)

Q_3 = 75th percentile (also called 3rd quartile)

Roughly divides the data into fourths

Measure of spread 2: Interquartile range (IQR) = $Q_3 - Q_1$

Box plots can also visualize quantitative data



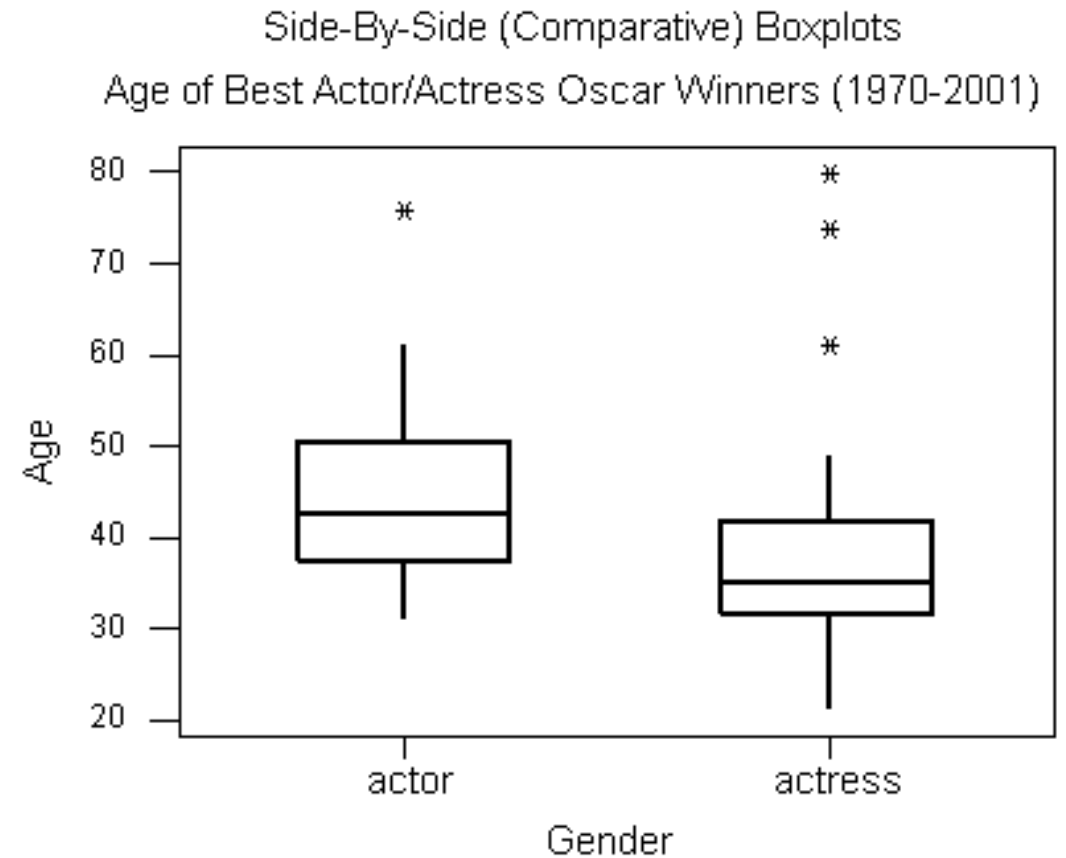
R: `boxplot(v)`

Side-by-side boxplots

Boxplots are particularly useful for comparing distributions!

Let's look at the ages that people won the best actor/actress Oscar

What does this figure tell us?



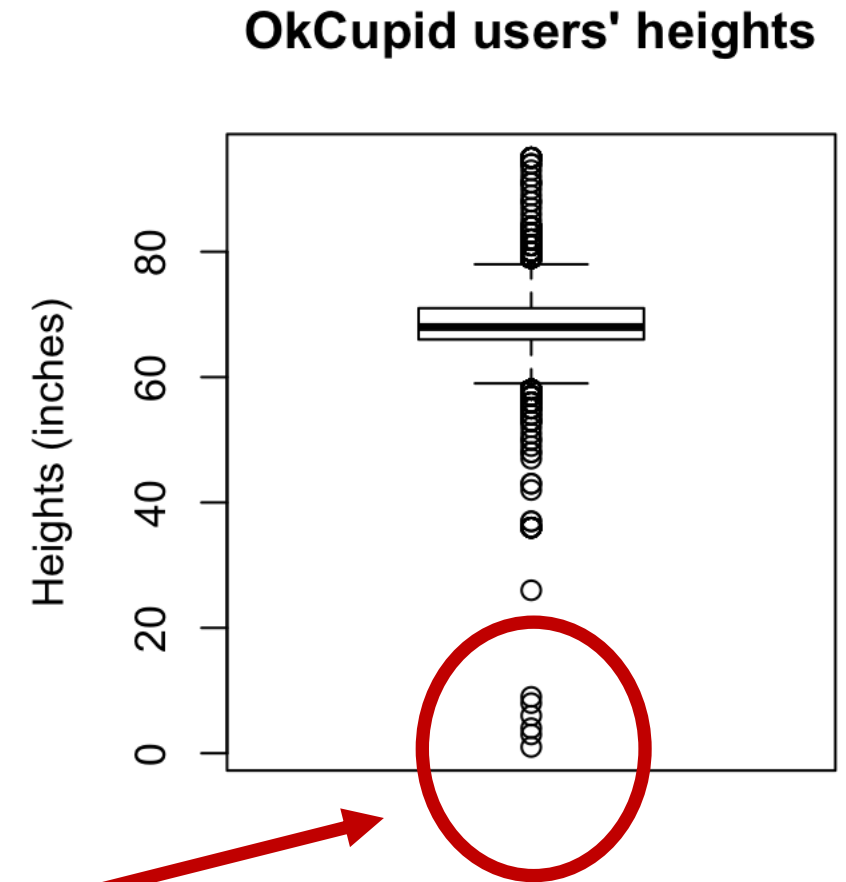
Outliers

Outliers on boxplots are values that are more than $1.5 * IQR$

What should we do if we have outliers?

Investigate!

- If they are due to an error, remove them



People under 20'' tall?

Outliers

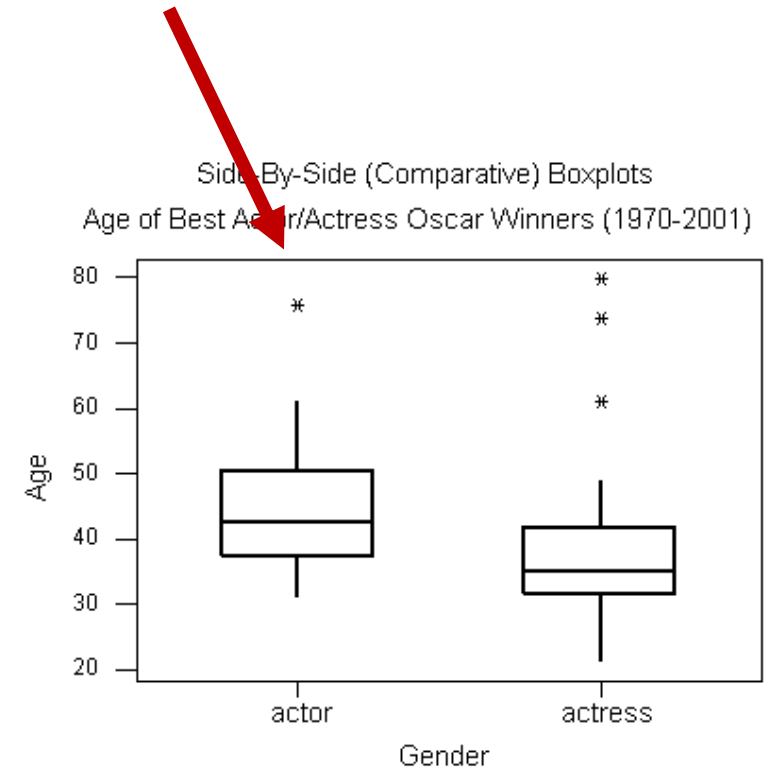
Outliers on boxplots are values that are more than $1.5 * IQR$

What should we do if we have outliers?

Investigate:

- If there are due to an error, remove them
- **If not, need to account for them**

Who is this actor?



Questions?



Let's try it in RStudio!

Visualizing two quantitative variables

CitiBike data

Let's look at the bike share data from NYC

```
load('daily_bike_totals.rda')
```



CitiBike analysis

	date	trips	precipitation	snow_depth	snowfall	max_temperature	min_temperature
1	2013-07-01	16650	0.8385830	0	0	77.00	71.96
2	2013-07-02	22745	0.0787402	0	0	82.04	71.96
3	2013-07-03	21864	0.5314960	0	0	82.94	73.04
4	2013-07-04	22326	0.0000000	0	0	87.08	75.02
5	2013-07-05	21842	0.0000000	0	0	89.96	75.92
6	2013-07-06	20467	0.0000000	0	0	91.94	78.08
7	2013-07-07	20477	0.0000000	0	0	91.94	78.08
8	2013-07-08	21615	0.2204720	0	0	89.06	73.04

What does each case correspond to?

Line plots

We can use the `plot(x, y)` function to create line plots

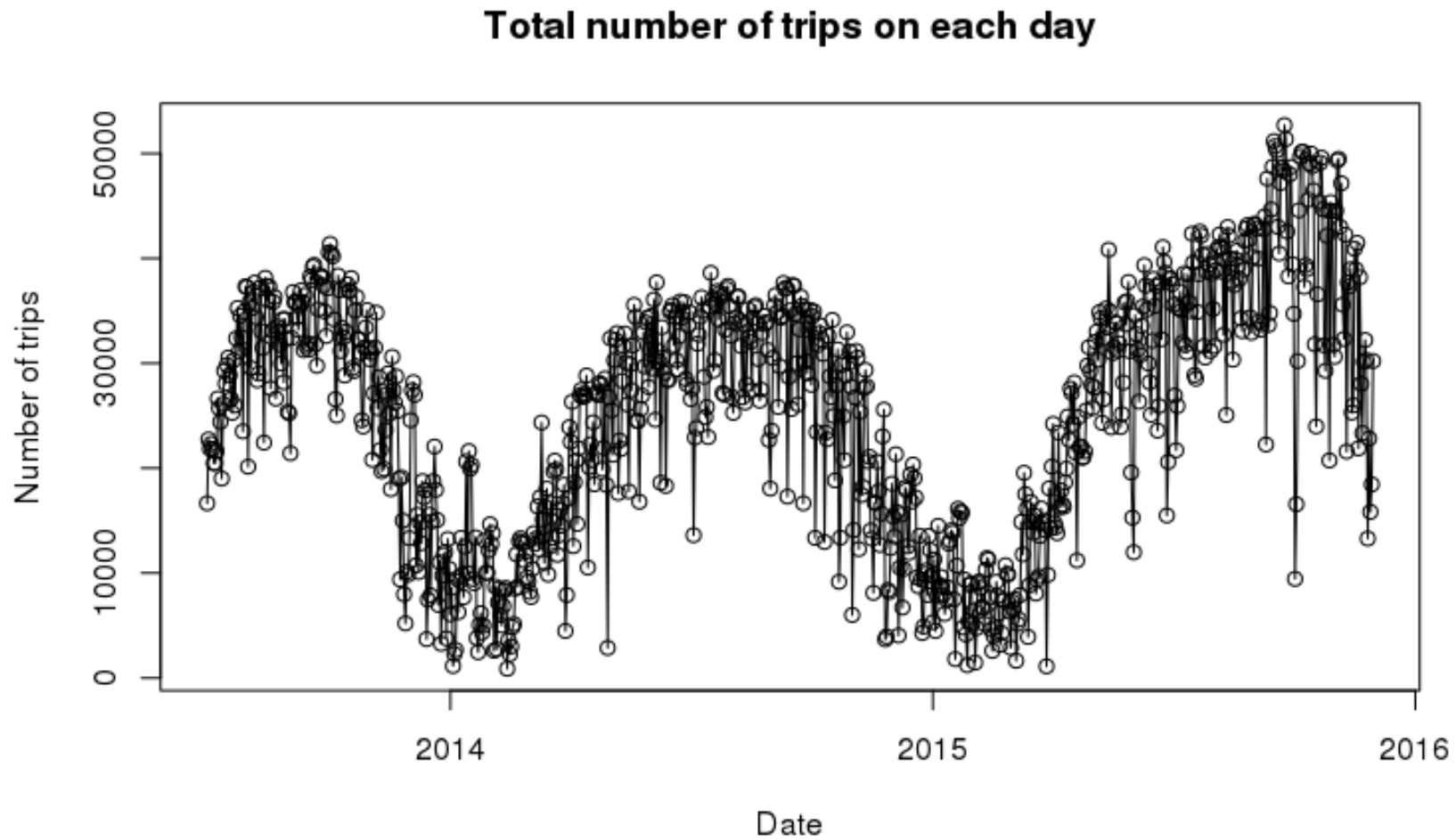
we can connect the points in a plot using

```
plot(x, y, type = 'l') # line graph
```

```
plot(x, y, type = 'o') # both points and a line
```

```
plot(bike_daily_data$date, bike_daily_data$trips,  
     type = 'o',  
     xlab = "Date",  
     ylab = "Number of trips",  
     main = "Total number of trips on each day")
```

Line plots

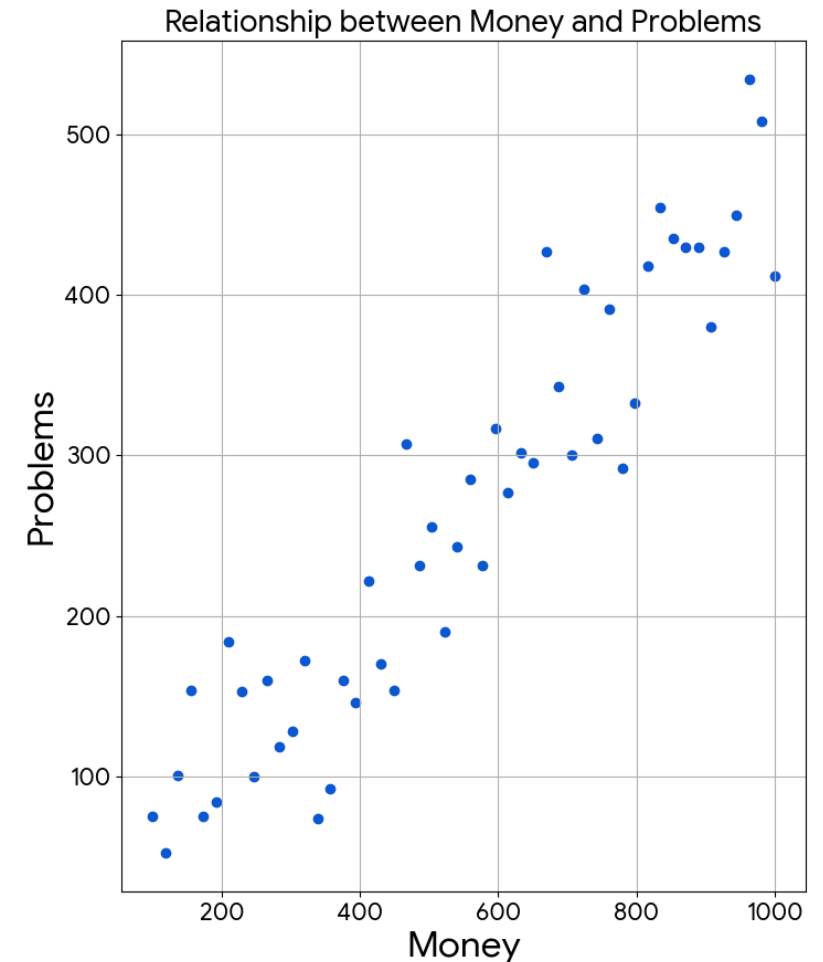


Scatterplot

A **scatterplot** graphs the relationship between two variables

- Each axis represents the value of one variables
- Each point the plot shows the value for the two variables for a single data case

If there is an explanatory and response variable, then the explanatory variable is put on the x-axis and the response variable is put on the y-axis



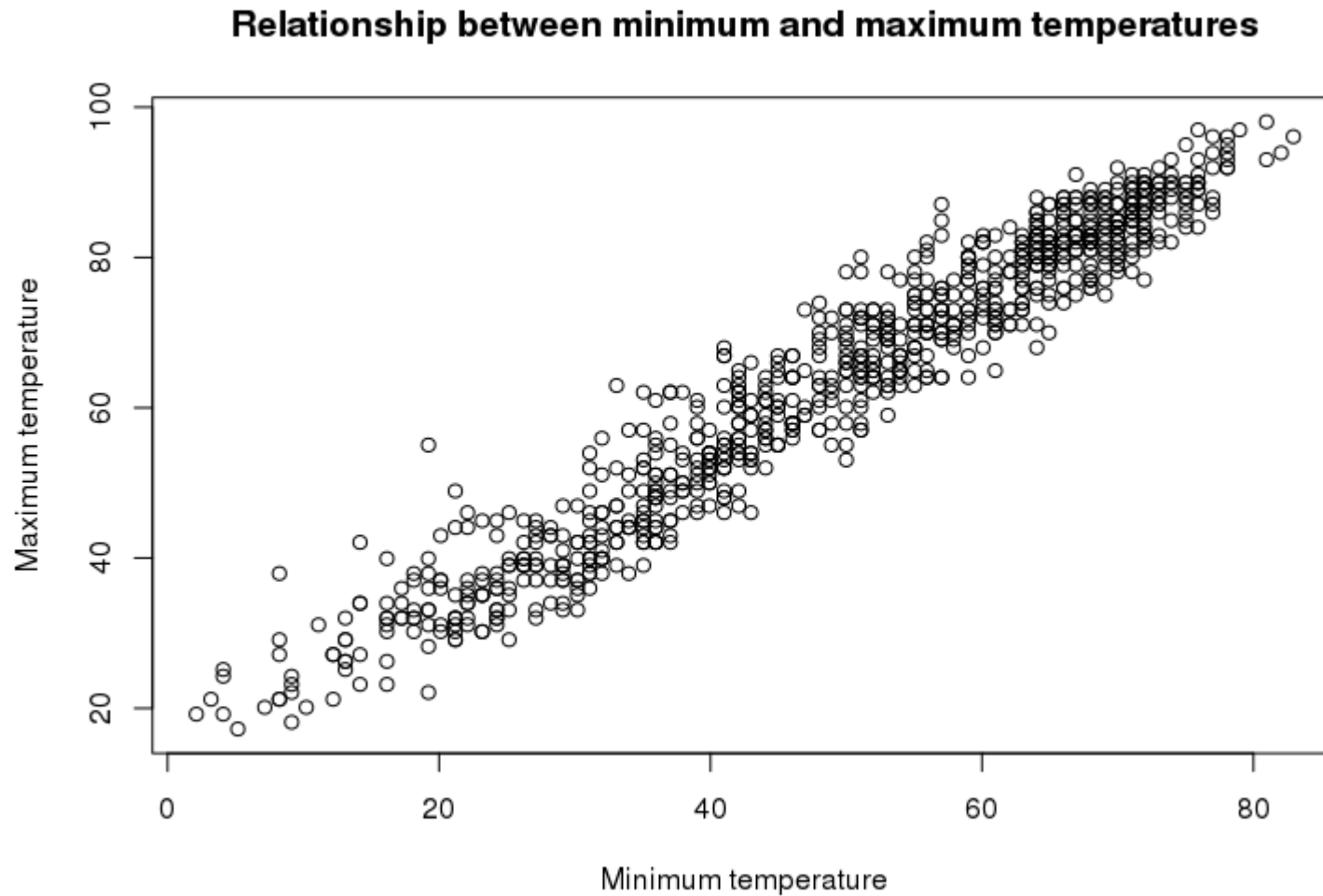
Scatter plots

We can use the `plot(x, y)` function to create scatter plots

Can you create a scatter plot of the relationship between the minimum and maximum temperatures?

```
plot(bike_daily_data$min_temperature,  
     bike_daily_data$max_temperature,  
     xlab = "Minimum temperature",  
     ylab = "Maximum temperature",  
     main = "Relationship between min and temp")
```

Scatter plots



The correlation coefficient

The **correlation** is statistic that measures of the strength and direction of a linear association between two variables

$$r = \frac{1}{(n - 1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

R: `cor(x, y)`

Properties of the correlation

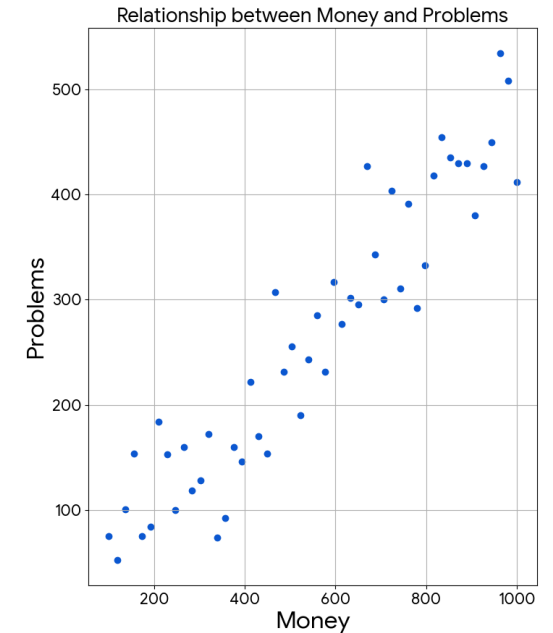
Correlation is always between -1 and 1: $-1 \leq r \leq 1$

The sign of r indicates the direction of the association

Values close to ± 1 show strong linear relationships, values close to 0 show no linear relationship

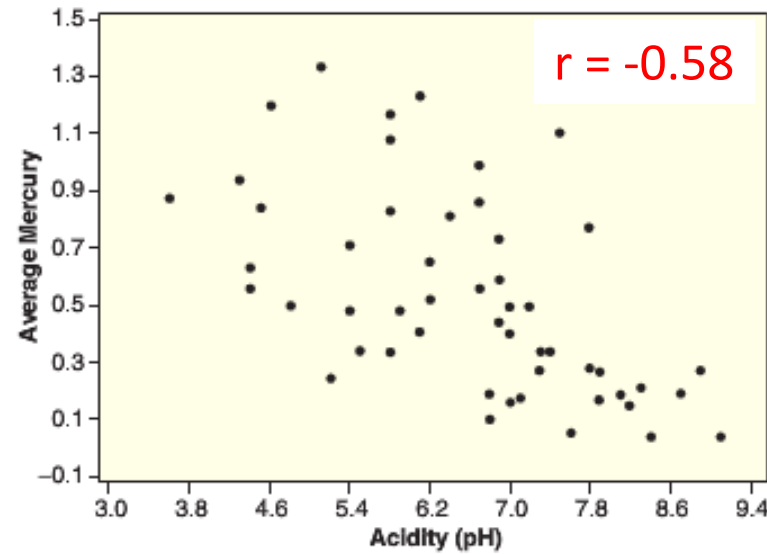
Correlation is symmetric: $r = \text{cor}(x, y) = \text{cor}(y, x)$

$$r = \frac{1}{(n-1)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s_x} \right) \left(\frac{y_i - \bar{y}}{s_y} \right)$$

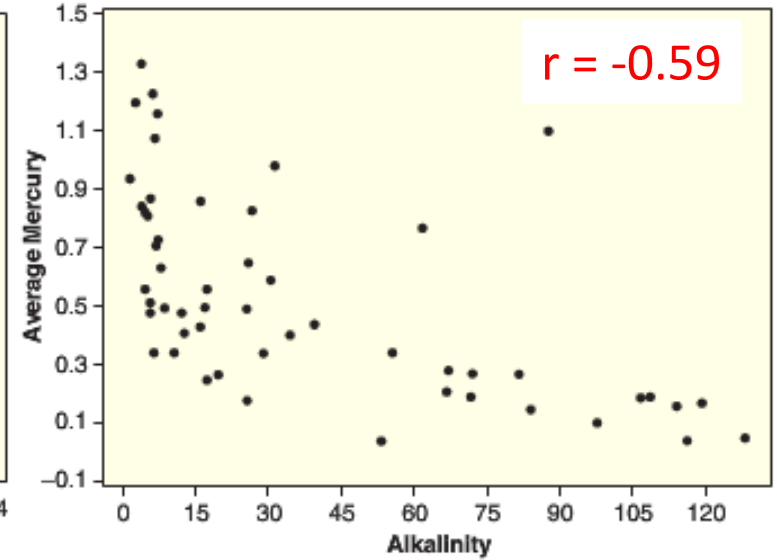


Florida lakes

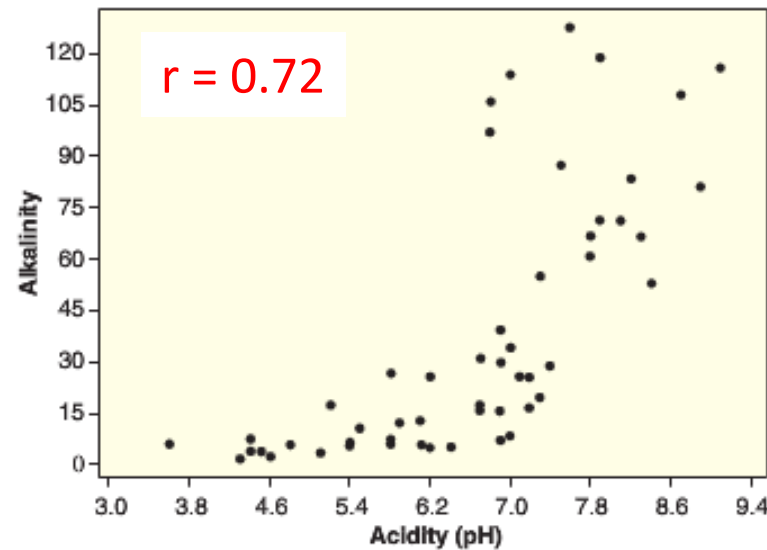
Correlation game



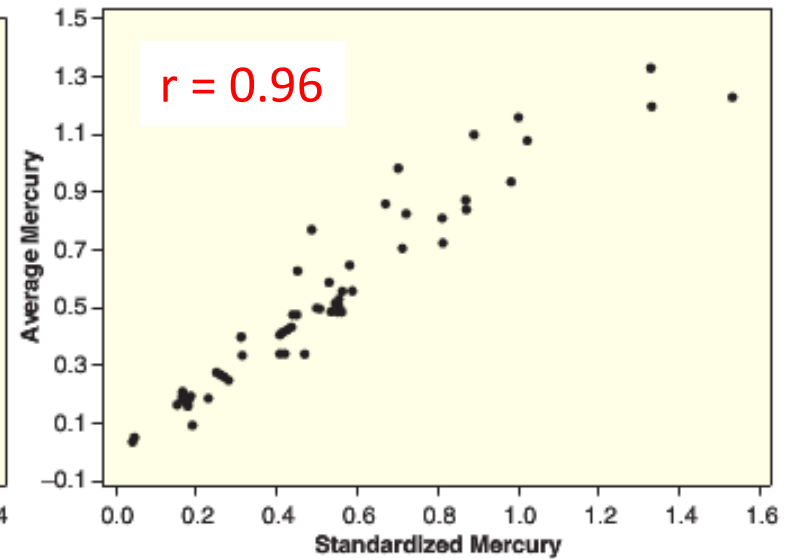
(a) Average mercury level vs acidity



(b) Average mercury level vs alkalinity



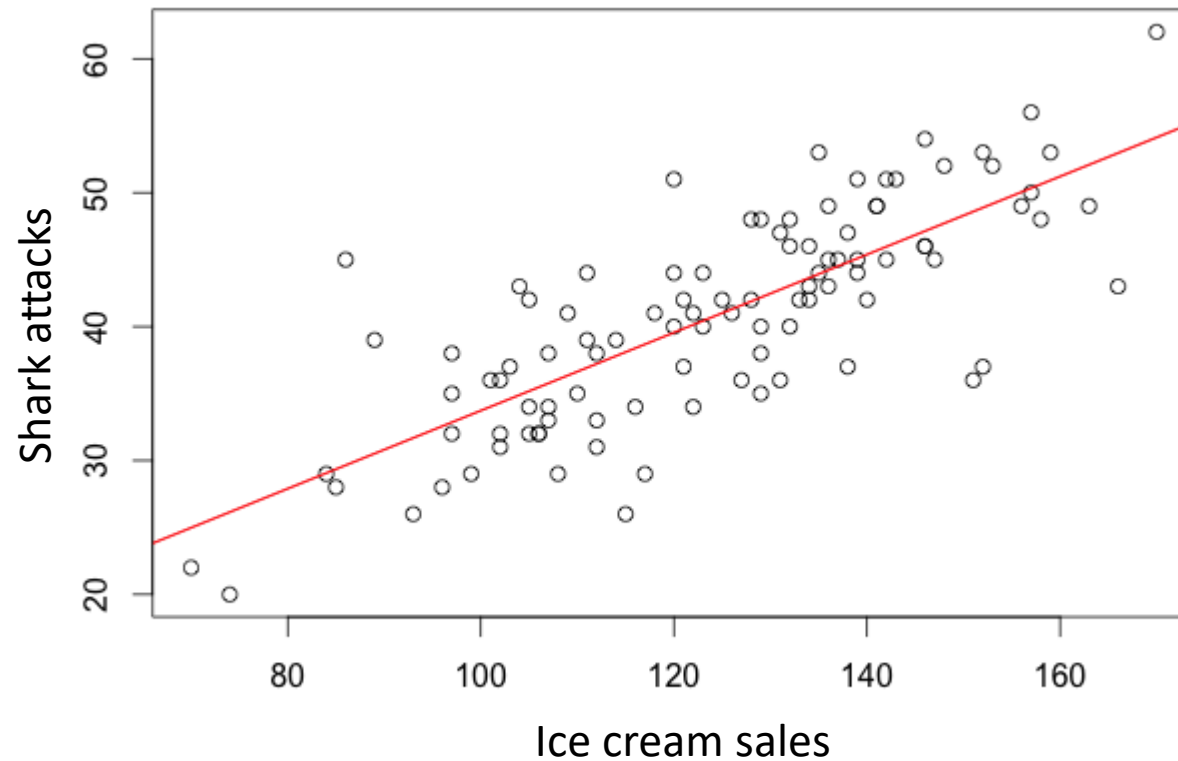
(c) Alkalinity vs acidity



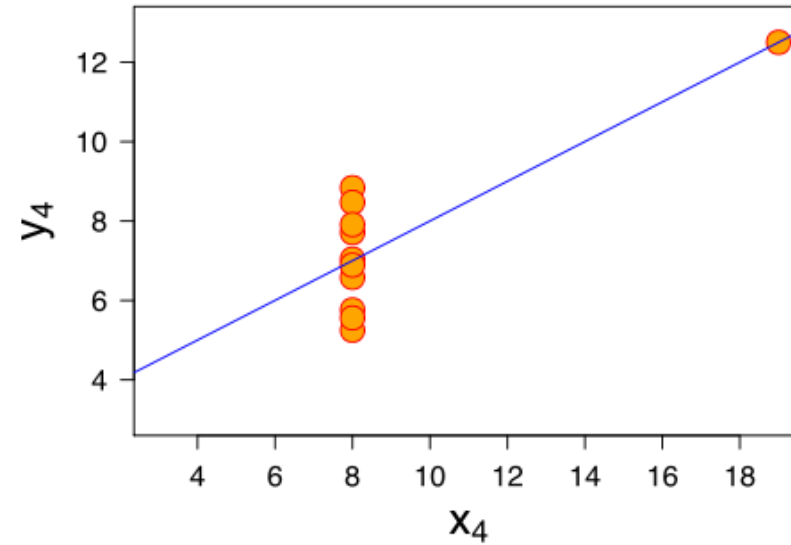
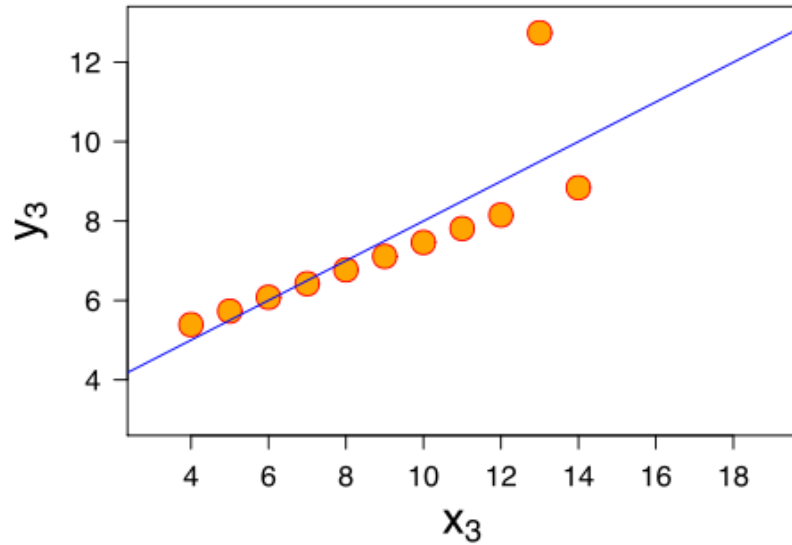
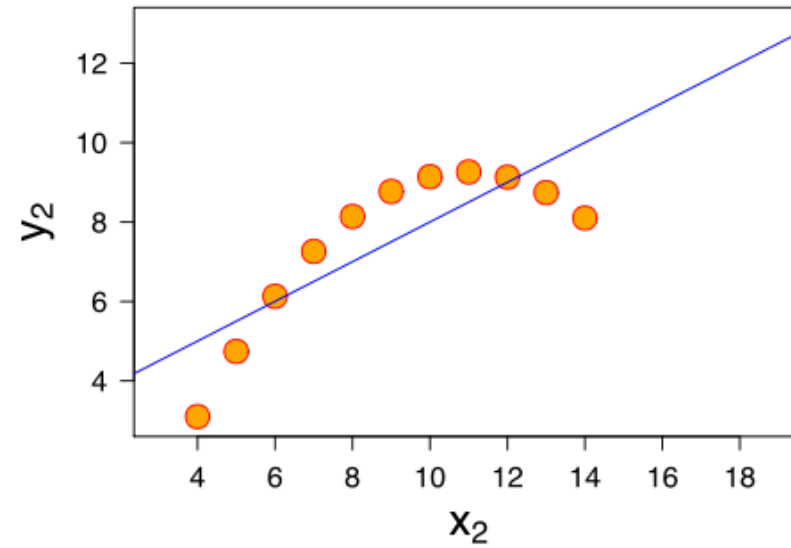
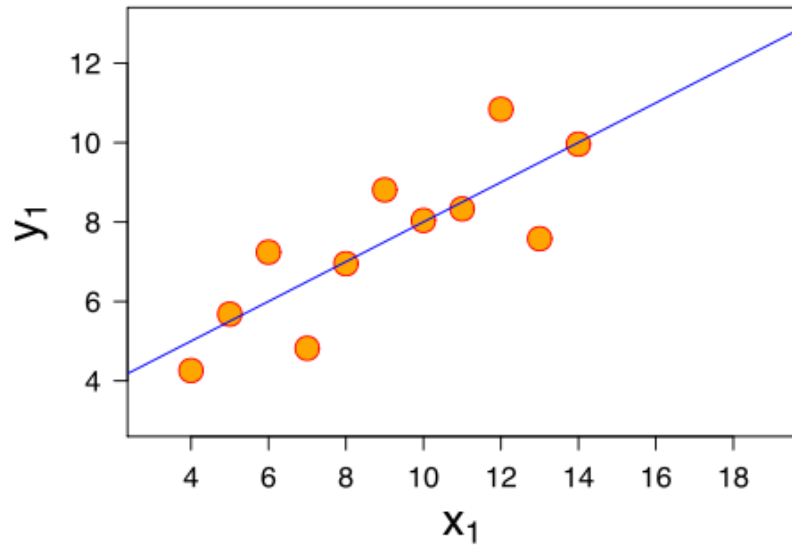
(d) Average vs standardized mercury levels

Correlation caution #1

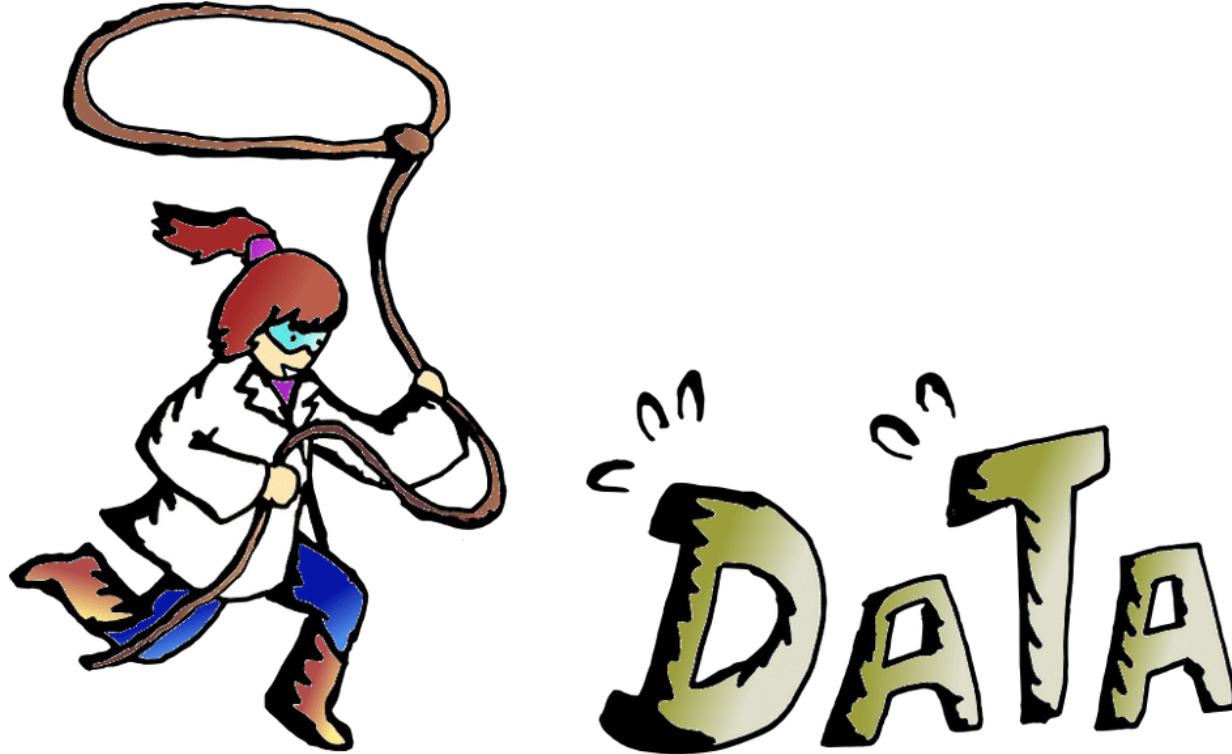
A strong positive or negative correlation does not (necessarily) imply a cause and effect relationship between two variables



Anscombe's quartet ($r = 0.81$)



Data wrangling/transformation using dplyr



The tidyverse and dplyr

The 'tidyverse'

The tidyverse is set of R packages that operate 'tidy data'

- i.e., that operate on data frames (or tibbles)

Tidy data is data where:

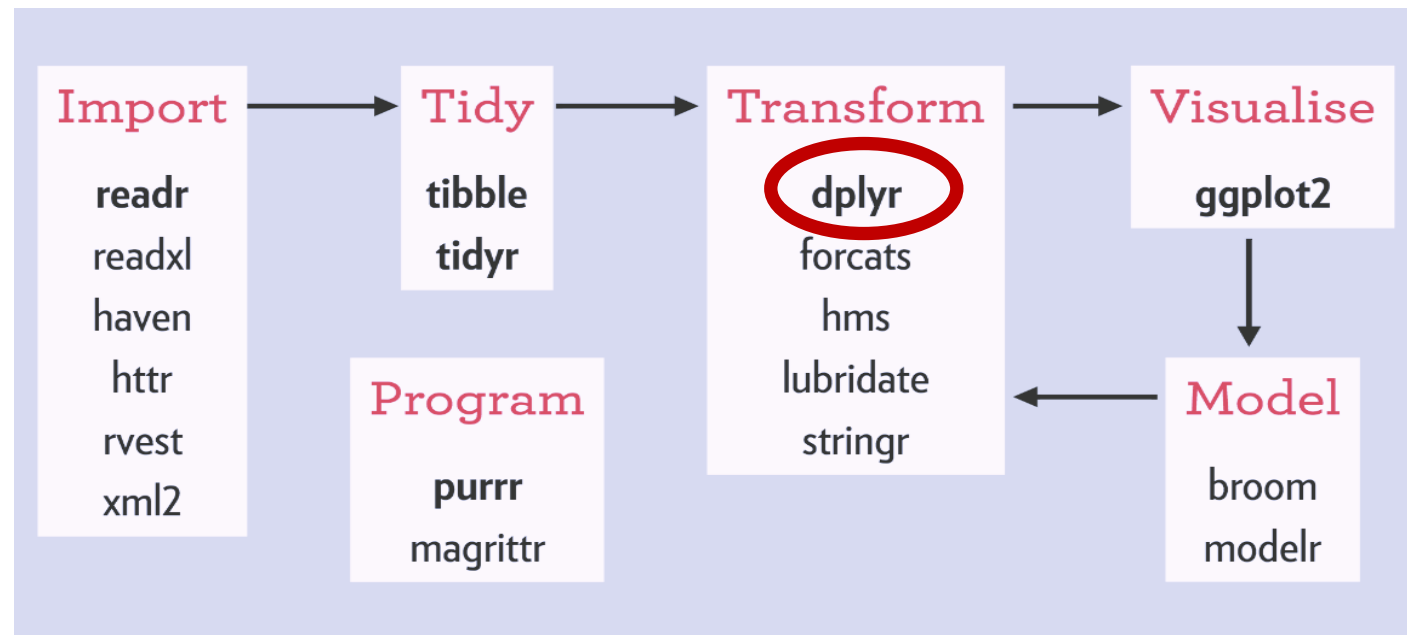
- Each variable must have its own column
- Each observation must have its own row
- Each value must have its own cell



The 'tidyverse'

The tidyverse is a set of packages share a common design philosophy

- Most written by Hadley Wickham



dplyr: A grammar for data wrangling

Grammar: a set of components that can be combined to achieve a goal

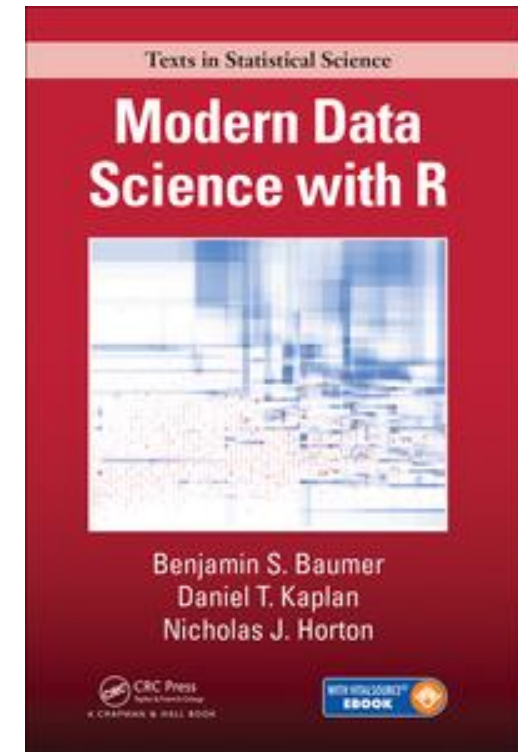
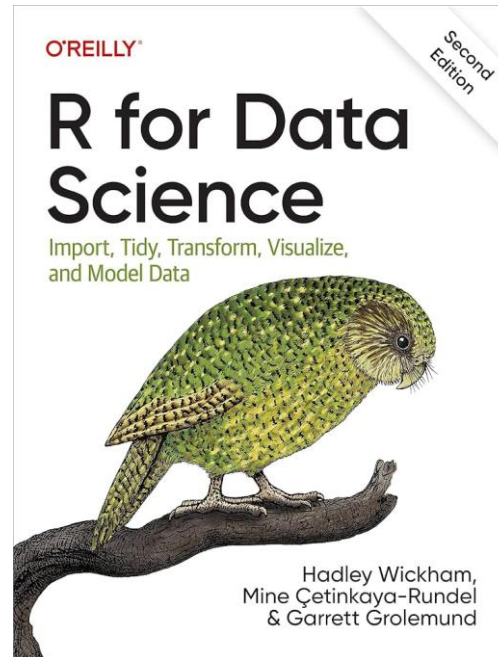
dplyr is a package that has a set of verbs that are useful for transformations data:

1. `filter()`
2. `select()`
3. `mutate()`
4. `arrange()`
5. `group_by()`
6. `summarize()`

All these function **take a data frame** and other arguments and **return a data frame**

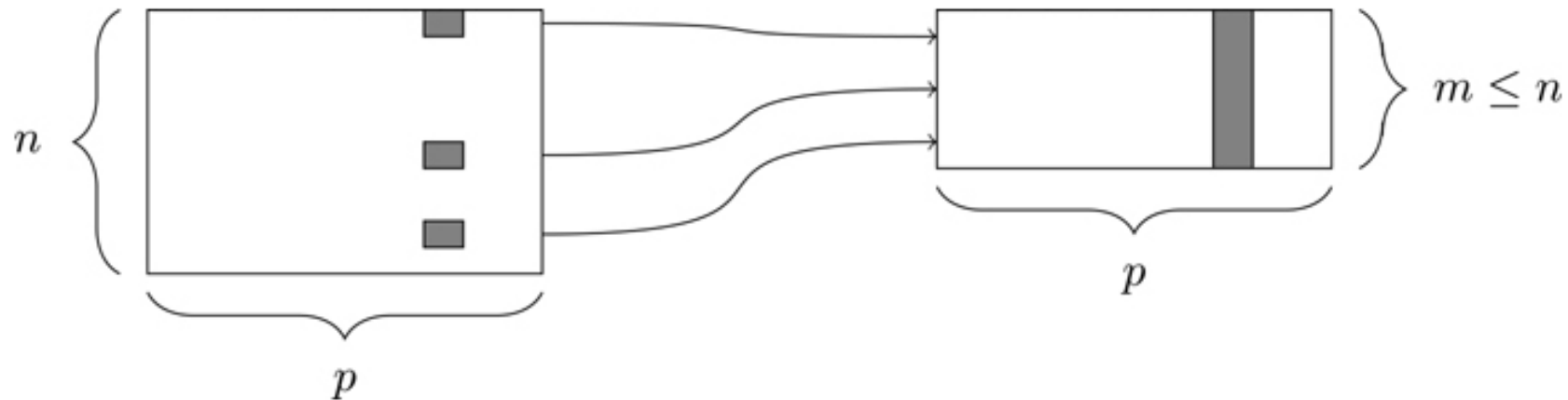
```
library(dplyr) # load the dplyr package
```

Quick overview of the dplyr functions



1. filter()

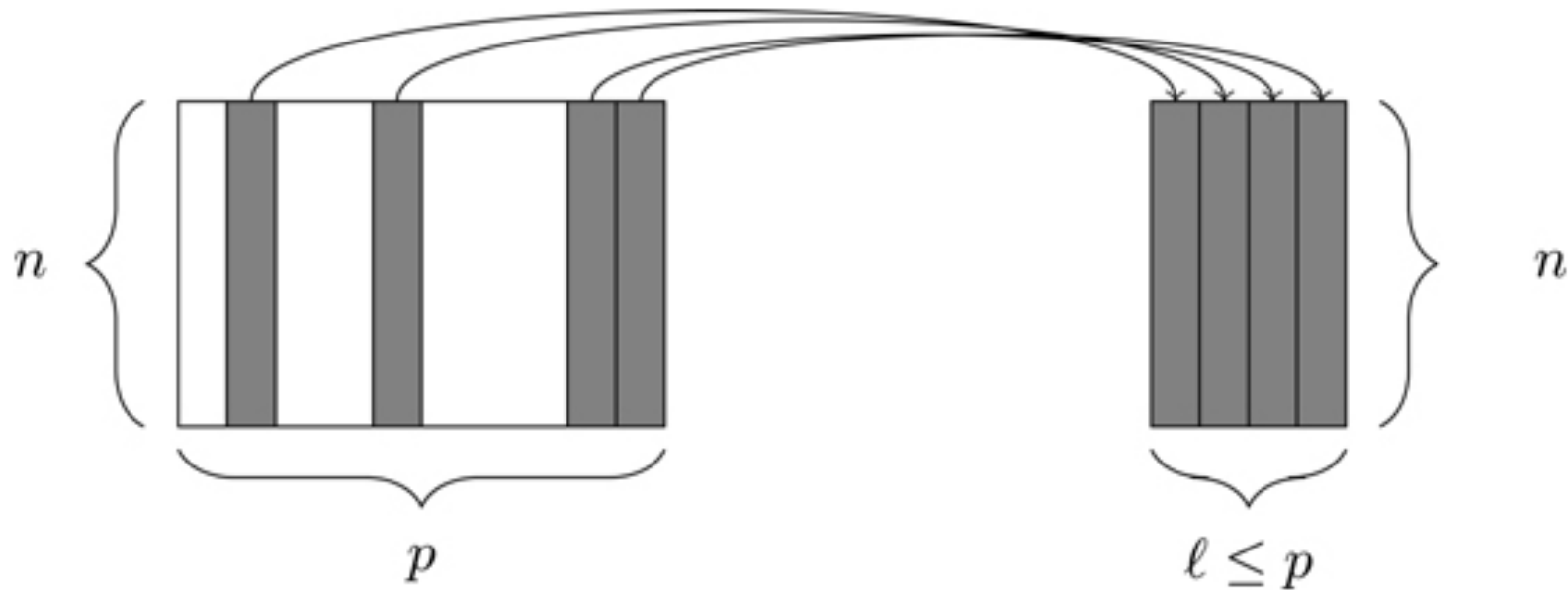
The `filter()` function allows you to select a subset of rows in data frame



```
filter(profiles, height == 77)
```

2. select()

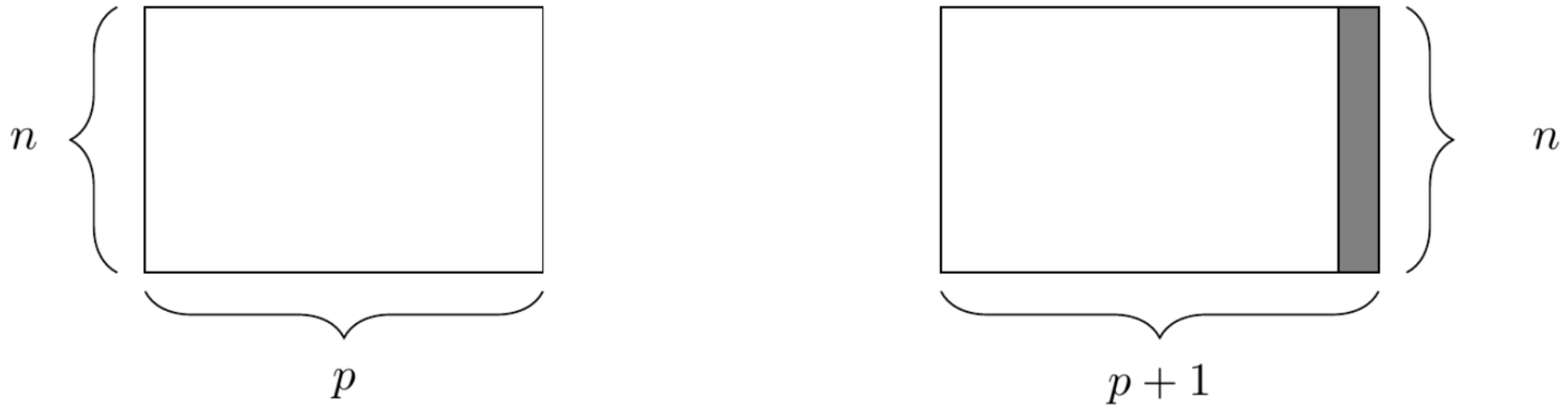
The `select()` function allows you to select a subset of columns



```
select(profiles, age, height)
```

3. mutate()

The `mutate()` function allows you to create new columns that are functions of existing columns

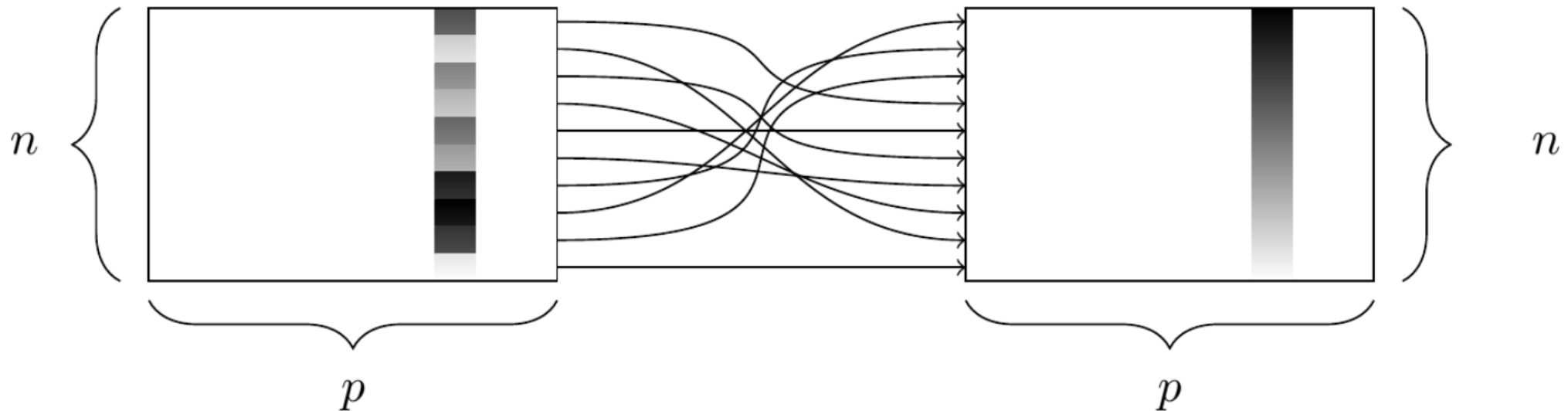


```
mutate(profiles, height_feet = height/12)
```

4. arrange()

The `arrange()` function arranges the rows based values in a column

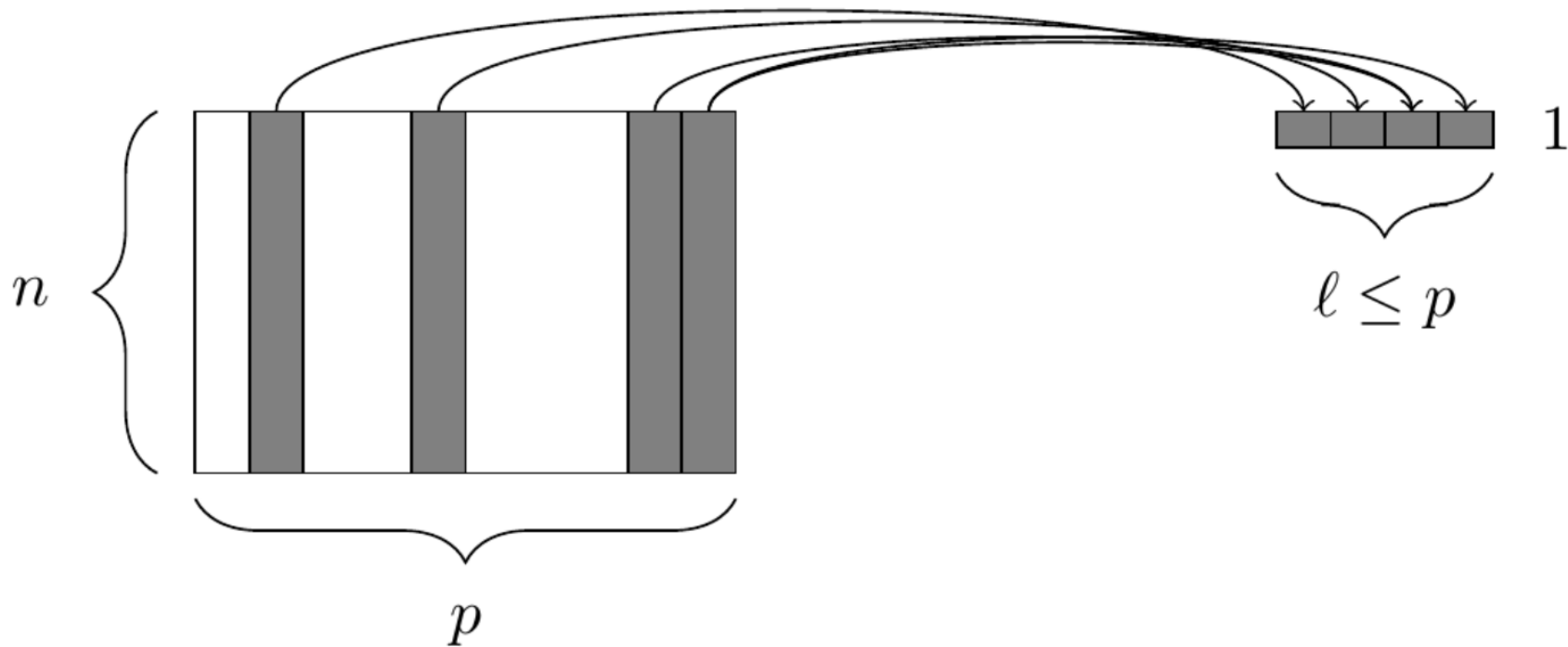
- `arrange(desc())` arranges from largest to smallest



`arrange(profiles, desc(height))`

5. summarize()

The `summarize()` function reduces values in many rows into single values

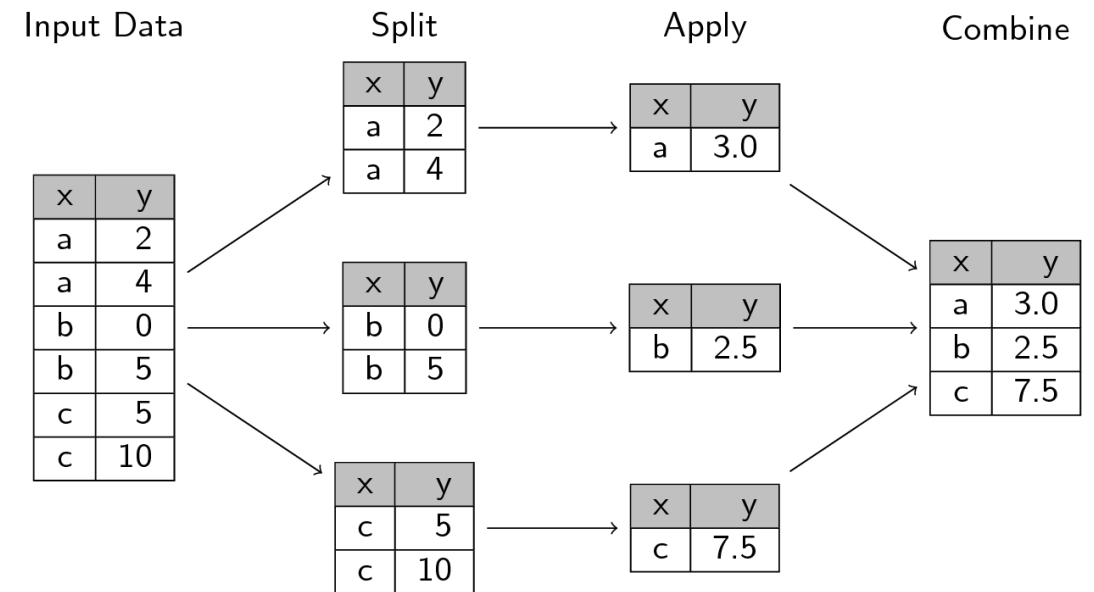
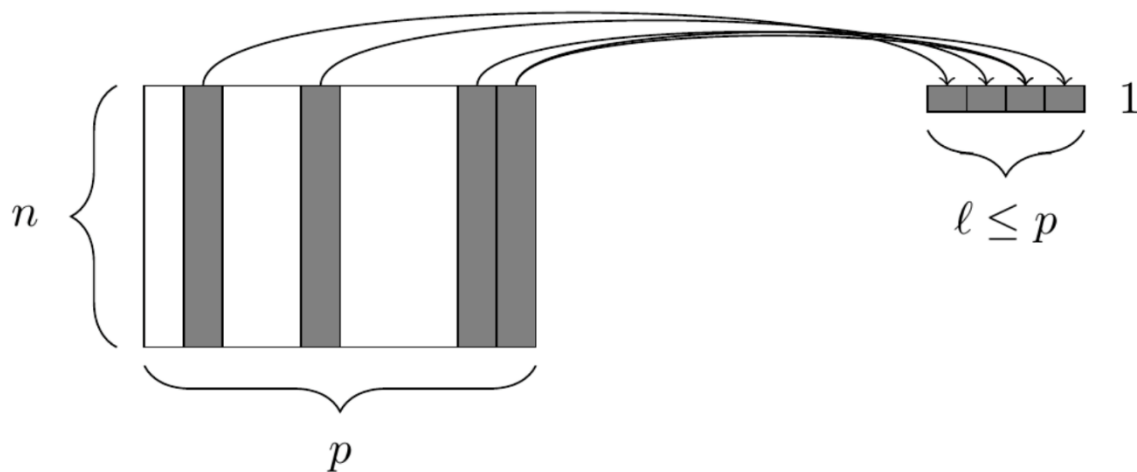


```
summarize(mean_age = mean(age))
```


6. The group_by() function

The `group_by()` function groups variables for future operations

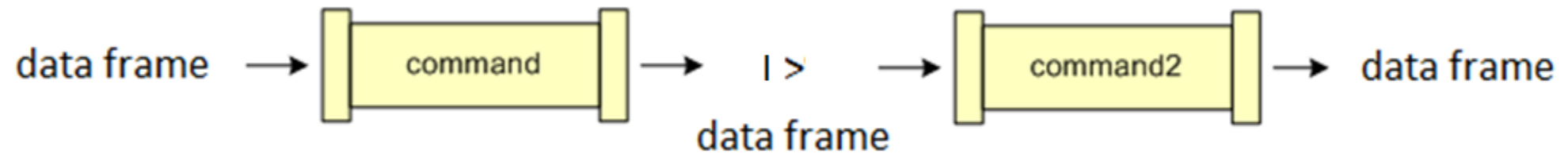
- It works in conjunction with `summarize()` and `mutate()` to do **split, apply, combine**



`group_by(profiles, drinks)`

The pipe operator

The pipe operator `|>` allows us to chain commands together



```
profiles|>
```

```
  group_by(sex) |>
```

```
  summarize(mean_age = mean(age))
```



Let's try it out!